

# LSSO: Solved Contextual Operators for Bidirectional Sequence Modeling

Zhaoyang Wang  
zhaoy@uir.edu.cn

## Abstract

Bidirectional sequence models can use the entire context to construct a compact state for global interaction. Recent extensions of recurrent memory and test-time training make this an increasingly useful alternative to dense attention, but leave open how to jointly design the state, its induced mixing, and its computational realization. We introduce LSSO, a contextual operator built around one compact direct solve per sample and head. A full-context cross-moment generates a strictly accretive system; its solved state is lifted to tokens through a QR soft frame and a learned scalar complement. The compact core covers every real strict contraction, including nonsymmetric and nonnormal maps, while the resulting token mixer has an exact gain computable in compact space. These guarantees concern the realized mixing operator, rather than the end-to-end input-dependent network. At fixed rank, arithmetic and activation storage scale linearly with sequence length. An algebraically equivalent CUDA implementation evaluates the solve and its analytic backward without a dense token map. In a matched three-seed comparison on eight GenomicBenchmarks tasks, LSSO obtains the highest mean accuracy on seven tasks and improves macro accuracy from the strongest alternative’s 75.57 to 78.53. Additional long-sequence results, trained-state diagnostics, and complete-mixer timings characterize the scope and cost of this full-context design.

## 1 Introduction

Image encoders and full-sequence classifiers receive all their tokens before producing a prediction. Each position can therefore draw on the entire sample, which makes global interaction a natural building block. Dense attention provides this interaction through token pairs, but its quadratic arithmetic becomes costly as the sequence grows. Compact-state alternatives offer another route: gather contextual information into a smaller representation and use it to transform every token [18, 22, 13]. The design of that representation determines both the global interaction and its computational cost.

Recurrent linear attention, selective state-space models, and test-time training (TTT) have developed powerful ways to construct such states from context [27, 9, 28]. Their extension to bidirectional tasks is now an active architectural direction. LION relates full, recurrent, and chunkwise forms of bidirectional linear attention [1]; ViT<sup>3</sup> studies full-batch inner updates for visual TTT [12]; and ZipMap compresses an image collection into a shared scene state for bidirectional reconstruction [14]. These approaches establish several ways to exploit complete context. In particular, bidirectional information access and parallel execution are distinct choices: a causal computation can be parallelized, and a bidirectional model can retain directional scans.

When the entire context is available, a layer can construct its state jointly from all tokens. This raises a design question: *what global transformation should the state induce, and how should its expressivity, evaluation, and gain be controlled?* The optimization perspective of test-time regression provides one framework for choosing a state [31], and MesaNet solves causal context regressions using conjugate gradients [30]. However, a smaller inner loss need not imply better task performance: recent analysis of KV-binding TTT motivates inspecting the induced operator itself [21]. We consequently treat the full-context operator as the design object, with its solve and geometry chosen together.

LSSO constructs this operator from a compact strictly accretive system. Each head gathers the full sample into a cross-moment  $Z = P^T C$ , generates a matrix  $K$  from  $Z$ , and obtains the contextual state by

**LSSO** A global operator generated from complete context

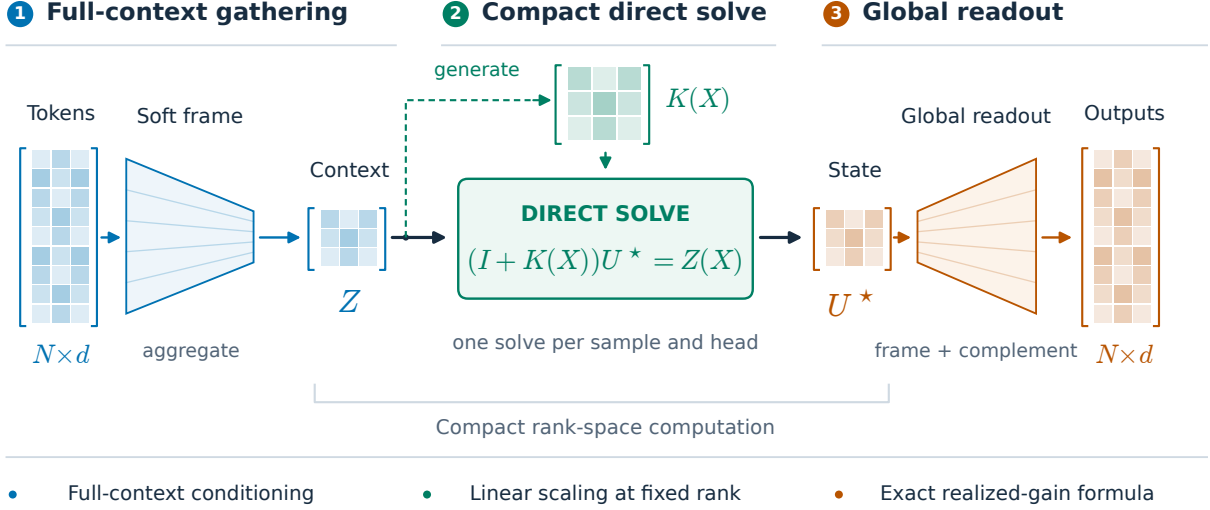


Figure 1: **Graphical abstract: full context, compact solve, global mixing.** All valid tokens jointly define a compact system. One direct solve per sample and head produces a shared state whose action reaches every output token. At fixed rank, the construction scales linearly with sequence length and admits an exact realized-gain formula. This formula describes the mixer after freezing its sample-conditioned quantities, not the end-to-end network Jacobian. The dashed path generates the system from the context statistic; the solid path evaluates it. Global readout uses the shared frame, context statistic, and scalar complement. Tensor shapes are schematic; no experimental results are depicted.

solving  $(\mathbf{I} + K)U^* = Z$ . The state is read back through the same soft frame  $P$  to produce a global token transformation (Figure 1). Equivalently, the compact action and its token-space lift are

$$M = 2(\mathbf{I} + K)^{-1} - \mathbf{I}, \quad (1)$$

$$T = \eta \mathbf{I} + P(M - \eta \mathbf{I})P^T. \quad (2)$$

All valid tokens participate in generating this shared operator. LSSO is a native full-context construction, rather than an equivalence transformation of an existing causal update rule.

The construction assigns a role to each structural choice. Accretivity makes the compact state uniquely solvable and controls the transposed solve used in backpropagation. Its reflected resolvent spans every strict contraction in *compact rank space*, so gain control still permits nonsymmetric mixing. The soft frame lifts this action to tokens, while  $\eta$  controls the complement outside the frame. Together they give an exact, sample-specific gain formula using matrices no larger than  $r \times r$ . Token-rank products and one small direct solve realize the operator with linear sequence scaling at fixed rank; the dense token matrix is never formed.

Our contributions are threefold:

1. **A solved full-context operator.** LSSO jointly constructs a compact state from all valid tokens and evaluates it by direct solution, without an inner learning rate, update schedule, or iterative stopping rule.
2. **Expressive compact geometry with a computable global gain.** The compact chart covers the real strict matrix ball, and its structured token-space lift admits an exact realized-gain certificate. A complementary monotonicity margin bounds the state and implicit adjoint.
3. **Task evidence and efficient realization.** Matched genomic experiments evaluate the complete mixer; LRA results and trained-state diagnostics examine further sequence settings. An equivalent native CUDA implementation demonstrates the execution cost of the compact construction.

Table 1: Representative full-context constructions. This comparison concerns how the state is formed, not a matched performance ranking.

| Construction               | Context representation                    | Evaluation route                               |
|----------------------------|-------------------------------------------|------------------------------------------------|
| LION [1]                   | Bidirectional linear-attention aggregates | Equivalent full, recurrent, or chunkwise forms |
| Visual full-batch TTT [12] | Updated inner-model parameters            | Full-context inner update and query evaluation |
| LSSO                       | Generated compact system and solved state | Direct compact solve and soft-frame readout    |

## 2 Related Work

**Global compact-state mixing.** Efficient Transformers replace the dense token map with projections, landmarks, kernels, cosine reweighting, or inducing states [33, 16, 5, 26, 35]. Set Transformer uses inducing points for global set interaction [18]; Luna packs and unpacks information through a fixed-length auxiliary sequence [22]; and ABC interprets several such mechanisms as learned bounded memory [25]. Perceiver similarly mediates global exchange through latent slots [13]. LSSO advances this compact-state line by building on XCiT’s central insight that an input-dependent cross-statistic can generate one compact global operator for all tokens [2]. XCiT turns token-aggregated cross-covariance into a shared channel mixer; LSSO carries the compact operator into token space, where the cross-moment generates an accretive system. Its equilibrium is solved directly, and the resulting contraction is lifted through a learned token frame without constructing a dense token matrix.

**From causal memory to full-context operators.** Fast-weight models and TTT construct a compact representation from a sequence [27, 28]. LION extends linear attention to bidirectional tasks through equivalent full, recurrent, and chunkwise forms [1]. ViT<sup>3</sup> studies full-batch visual inner updates [12], while ZipMap uses TTT to form a shared state from an image collection [14]. These are distinct ways to make complete context available to the output. LSSO contributes a different state construction: a full-context statistic generates a compact system whose direct solution defines the global mixing map. The available experiments do not yet compare LSSO against LION or visual TTT under a matched training protocol.

**Context as a solved problem.** Longhorn and test-time regression relate sequence processing to regression objectives [20, 31]. MesaNet optimizes causal per-context regressions with a chunkwise conjugate-gradient formulation [30]. Our bidirectional construction instead generates one compact accretive problem per sample and head. Its mathematical solution is evaluated with a direct factorization, subject to floating-point error. Analysis of KV-binding TTT further shows that inner optimization and task performance need not track one another in the studied settings [21]. This motivates our focus on the resulting operator; we do not infer a task advantage merely from solving a system accurately.

**Context-conditioned contractive geometry.** S4 and selective state-space models obtain linear sequence scaling through structured recurrence and input-dependent state evolution [10, 9, 7]. Deep equilibrium models use implicit differentiation at a root [3]; monotone operator equilibrium networks additionally obtain well-posed states from monotone structure [34]. Direct Cayley parameterization gives a complete trainable chart for static Lipschitz-bounded network weights [32]. LSSO makes this geometry contextual: every sample generates a compact accretive operator, its exact solve produces the global token mixer, and compact statistics recover the gain actually realized on that sample. Skew coordinates retain nonsymmetric and nonnormal contractions. This joins global noncausal conditioning with accretive/resolvent geometry [4] and provides a directly measurable operator guarantee in a setting where ordinary self-attention need not be globally Lipschitz [17].

**Hardware-efficient linear layers.** FlashAttention demonstrated that avoiding materialized intermediates and controlling memory traffic can matter as much as operation count [6]; Mamba likewise couples its

recurrence to a hardware-aware parallel algorithm [9]. LSSO follows this algorithm–implementation co-design principle. Its native path lowers the compact algebra to fused CUDA schedules for both the forward and analytic backward. Unlike exact softmax attention, whose pairwise arithmetic remains quadratic even when FlashAttention avoids storing the score matrix, LSSO’s compact algebra is linear in sequence length at fixed rank. Section 6 reports its measured behavior, and Appendix D gives the lowering and workspace accounting.

### 3 LSSO

We describe one head and omit the batch index. Let  $X \in \mathbb{R}^{N \times D}$  be the incoming token matrix,  $d$  the head width, and  $r$  the compact rank. A single input projection produces relation and content coordinates

$$[A, C] = XW_{bc} + b_{bc}, \quad A \in \mathbb{R}^{N \times r}, \quad C \in \mathbb{R}^{N \times d}. \quad (3)$$

Heads are concatenated and passed through an ordinary output projection after mixing. The construction below is repeated independently for each head, with one compact state shared by all valid tokens in that head.

#### 3.1 Full-context computation at a glance

For normalized, position-transformed relation coordinates, LSSO builds a soft frame  $P$  and evaluates

$$Z = P^\top C, \quad U^* = (\mathbf{I}_r + K(Z))^{-1}Z, \quad Y = \eta C + P[2U^* - (1 + \eta)Z]. \quad (4)$$

The first operation gathers the entire valid context; the second constructs a shared solved state; the third returns its action to all tokens. The following subsections specify each operation. Masking,  $1/\sqrt{n}$  relation normalization, and Rank-Rotary precede frame construction, as detailed in Section 3.5; throughout,  $n$  is the valid-token count.

#### 3.2 Gathering the full context into a compact statistic

The QR soft frame maps unconstrained relation coordinates to a strict column contraction. Form the reduced QR factorization

$$\begin{bmatrix} P \\ E \end{bmatrix} R_{\text{qr}} = \begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix}, \quad \begin{bmatrix} P \\ E \end{bmatrix}^\top \begin{bmatrix} P \\ E \end{bmatrix} = \mathbf{I}_r, \quad (5)$$

where  $P \in \mathbb{R}^{N \times r}$  is the token-space frame. A common positive rescaling, which leaves the exact  $Q$  factor unchanged, controls the numerical range. The identity augmentation means that the factorization is defined even when  $N < r$  or  $A$  is rank deficient.

The frame compresses the content into one shared cross-moment,

$$Z = P^\top C \in \mathbb{R}^{r \times d}. \quad (6)$$

$Z$  aggregates all valid positions symmetrically in one noncausal pass without forming the token-level matrix in (2).

#### 3.3 Generating and solving the contextual core

In the default DYNAMIC mode,  $Z$  generates compact raw coordinates

$$\Theta = \Theta_0 + \frac{ZW_{\text{drive}}}{\sqrt{n}}, \quad (7)$$

where  $\Theta_0 \in \mathbb{R}^{r \times r}$  and  $W_{\text{drive}} \in \mathbb{R}^{d \times r}$  are learned per-head parameters. The STATIC ablation uses  $\Theta = \Theta_0$ , while ZERO fixes the compact contraction to zero.

For DYNAMIC and STATIC, define

$$L(\Theta) = \text{tril}(\Theta, -1) + \text{Diag}(\text{softplus}(\text{diag}(\Theta) + c_1)), \quad (8)$$

$$\Omega(\Theta) = \text{triu}(\Theta, 1) - \text{triu}(\Theta, 1)^\top, \quad (9)$$

$$K(\Theta) = L(\Theta)L(\Theta)^\top + \Omega(\Theta), \quad (10)$$

where  $c_1 = \text{softplus}^{-1}(1)$ . Thus  $L$  is lower triangular with positive diagonal, while  $\Omega$  is skew-symmetric. The compact equilibrium and the pre-output token coordinates are

$$U^* = (\mathbf{I}_r + K)^{-1}Z, \quad (11)$$

$$Y = \eta C + P[2U^* - (1 + \eta)Z], \quad \eta = \tanh(\eta_{\text{raw}}) \in (-1, 1). \quad (12)$$

The complement is learned independently for each head and is initialized to 0.9. In ZERO,  $M = 0$ , which gives

$$Y = \eta(C - PZ). \quad (13)$$

At  $\Theta = 0$ ,  $L = \mathbf{I}_r$ ,  $\Omega = 0$ ,  $K = \mathbf{I}_r$ , and the compact contraction is  $M = 0$ . Both nonzero-core modes start at that point; DYNAMIC also starts with  $W_{\text{drive}} = 0$ .

### 3.4 Lifting the compact action to all tokens

Define the reflected resolvent

$$M = 2(\mathbf{I}_r + K)^{-1} - \mathbf{I}_r = (\mathbf{I}_r - K)(\mathbf{I}_r + K)^{-1}. \quad (14)$$

Substituting  $Z = P^\top C$  into (12) gives the exact token-mixing form

$$Y = TC, \quad T = \eta \mathbf{I}_N + P(M - \eta \mathbf{I}_r)P^\top. \quad (15)$$

**Solved contextual inner state.** LSSO obtains  $U^*$  by a direct  $r \times r$  solve. Appendix A.3 relates this solution to a globally convergent contextual dynamics.

### 3.5 Position, masking, and execution order

For a boolean validity mask  $m \in \{0, 1\}^N$ , invalid rows of both  $A$  and  $C$  are set to zero before any compact statistic is formed. Let

$$n = \sum_{i=1}^N m_i. \quad (16)$$

For a nonempty sequence, relation coordinates are normalized as  $A \leftarrow A/\sqrt{n}$ . An entirely masked sequence is assigned the zero output by definition. Masking before normalization makes the result invariant to appended padding.

Rank-Rotary applies a block-diagonal planar rotation to the  $r$  relation coordinates before the soft frame. For centered position  $p_i$  and frequencies  $\{\omega_j\}_{j=1}^{r/2}$ , each coordinate pair uses the angle  $p_i \omega_j$ . The transformation preserves each relation row's Euclidean norm and obeys the relative-phase identity

$$\mathcal{R}(p_i)^\top \mathcal{R}(p_j) = \mathcal{R}(p_j - p_i). \quad (17)$$

Rank-Rotary therefore supplies relative phase within relation rank space while preserving each row norm. Absolute or two-dimensional position embeddings remain external to the mixer. Full-context access does not remove positional structure: these feature rotations are applied before aggregation and do not impose a causal visibility mask.

### 3.6 Linear sequence complexity

For one head, frame formation costs  $O(Nr^2)$ , the two frame-content contractions cost  $O(Nrd)$ , and construction and solution of the compact problem cost  $O(r^2d + r^3)$ . The forward arithmetic and activation/workspace storage are therefore

$$\mathcal{C}_{\text{LSSO}} = O(Nr^2 + Nrd + r^2d + r^3), \quad (18)$$

$$\mathcal{M}_{\text{LSSO}} = O(N(r + d) + r^2 + rd). \quad (19)$$

The analytic backward reuses compact factors and performs the same classes of token-rank contractions, so its arithmetic and saved-state dependence on  $N$  is also linear. Multiplying the head-local terms by the number of heads does not change the sequence-length degree.

Unlike exact MHA, whose  $O(N^2d)$  mixing cost persists under FlashAttention [6], LSSO has  $O(Nr(r + d))$  token-dependent work and no  $N \times N$  intermediate. Both retain  $O(ND^2)$  feature projections.

## 4 Geometry of the Full-Context Operator

We analyze one realized head in exact arithmetic. Complete proofs are given in Appendix A, and mixed-precision execution is described in Section 6.

### 4.1 Is the contextual state well posed?

**Theorem 4.1** (Stable solved state and implicit adjoint). *Define the per-sample monotonicity margin*

$$\mu = 1 + \sigma_{\min}(L)^2 > 1, \quad A_K = \mathbf{I}_r + K. \quad (20)$$

Then  $U^\star = A_K^{-1}Z$  exists uniquely, and

$$\sigma_{\min}(A_K) \geq \mu, \quad \|U^\star\|_F \leq \mu^{-1} \|Z\|_F, \quad \|A_K^{-\top} G\|_F \leq \mu^{-1} \|G\|_F. \quad (21)$$

For a second generated problem  $A'_K = \mathbf{I}_r + K'$  with margin  $\mu'$  and solution  $U^{\star'} = A'^{-1}_K Z'$ , the finite contextual perturbation obeys

$$\|U^\star - U^{\star'}\|_F \leq \mu^{-1} \|Z - Z'\|_F + (\mu\mu')^{-1} \|K - K'\|_2 \|Z'\|_F. \quad (22)$$

Consequently, an infinitesimal perturbation obeys

$$\|dU^\star\|_F \leq \mu^{-1} \|dZ\|_F + \mu^{-2} \|dK\|_2 \|Z\|_F. \quad (23)$$

The same computable margin controls the forward state, the transposed solve used by implicit differentiation, and sensitivity to the generated compact problem. The realized-gain certificate below separately measures the contraction margin, which also depends on the skew coordinates.

### 4.2 What does gain control allow the operator to express?

**A frame that preserves the compact guarantee.**

**Theorem 4.2** (Soft-frame contract). *Let  $P$  be the upper block of the  $Q$  factor in (5). If  $\sigma_i(A)$  are the singular values of the normalized relation matrix, then*

$$\sigma_i(P) = \frac{\sigma_i(A)}{\sqrt{1 + \sigma_i(A)^2}}. \quad (24)$$

Consequently,

$$P^\top P \prec \mathbf{I}_r, \quad \mathbf{I}_N - PP^\top \succ 0, \quad 1 - \|P\|_2^2 = \frac{1}{1 + \|A\|_2^2}. \quad (25)$$

Moreover, at the level of represented frames, the construction covers every  $P \in \mathbb{R}^{N \times r}$  satisfying  $P^\top P \prec \mathbf{I}_r$ .

*Proof sketch.* The lower block of (5) obeys  $ER_{\text{qr}} = \mathbf{I}_r$ . Since the augmented matrix has full column rank,  $R_{\text{qr}}$  and hence  $E$  are nonsingular. Orthogonality of the full thin- $Q$  factor gives

$$P^\top P + E^\top E = \mathbf{I}_r. \quad (26)$$

Because  $E^\top E \succ 0$ , the first claim follows. Also  $R_{\text{qr}}^\top R_{\text{qr}} = \mathbf{I}_r + A^\top A$  and  $P = AR_{\text{qr}}^{-1}$ , which yields (24) and the exact norm margin. In particular,  $\|P\|_2 < 1$ , which implies  $PP^\top \prec \mathbf{I}_N$  and proves the second claim.  $\square$

The identity augmentation supplies the strict complement  $\mathbf{I}_N - PP^\top$ , which allows the compact contraction to extend to every token direction. If  $A_0$  denotes the masked relation matrix before division by  $\sqrt{n}$ , the final term in (25) becomes  $1/(1 + \|A_0\|_2^2/n)$ . Thus bounded per-token relation energy prevents the frame margin from deteriorating merely because the sequence is longer.

**Proposition 4.3** (Length-normalized contextual drive). *For the dynamic core,*

$$\|Z\|_F \leq \|P\|_2 \|C\|_F, \quad \|\Theta - \Theta_0\|_F \leq \frac{\|C\|_F}{\sqrt{n}} \|W_{\text{drive}}\|_2. \quad (27)$$

*The first inequality is strict relative to  $\|C\|_F$  whenever  $C \neq 0$ . Accordingly, if  $\|C\|_F^2/n \leq c^2$ , the magnitude of the contextual drive is at most  $c\|W_{\text{drive}}\|_2$ , independently of sequence length.*

#### A complete compact contraction chart.

**Definition 4.4** (Strict accretivity). *A real square matrix  $K$  is strictly accretive when*

$$\text{sym}(K) := \frac{K + K^\top}{2} \succ 0. \quad (28)$$

The Cayley connection between accretive coordinates and contractions is classical in monotone operator theory [4], and direct Cayley charts have parameterized static Lipschitz-bounded network weights [32]. The result below gives the complete strict form needed by LSSO. Its role here is contextual:  $K$  is generated from the current sample's cross-moment, and the resulting contraction is lifted through that sample's token frame.

**Theorem 4.5** (Accretive Cayley parameterization). *The map in (10) ranges over all real strictly accretive  $r \times r$  matrices. For every such  $K$ , the reflected resolvent  $M$  in (14) is a strict spectral-norm contraction and obeys*

$$\mathbf{I}_r - M^\top M = 4(\mathbf{I}_r + K)^{-\top} LL^\top (\mathbf{I}_r + K)^{-1} \succ 0. \quad (29)$$

*Conversely, the map  $K \mapsto M$  is a bijection from real strictly accretive matrices to the open real matrix ball  $\{M : \|M\|_2 < 1\}$ , with inverse*

$$K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}. \quad (30)$$

*Proof sketch.* The construction has  $\text{sym}(K) = LL^\top \succ 0$ , because  $L$  has positive diagonal. Conversely, for any strictly accretive  $K$ , the Cholesky factor of  $\text{sym}(K)$  and the skew remainder  $K - \text{sym}(K)$  recover  $L$  and  $\Omega$ , proving completeness.

Let  $Q = \mathbf{I}_r + K$ . Strict accretivity implies that  $Q$  is nonsingular. Using  $M = (\mathbf{I}_r - K)Q^{-1}$  gives

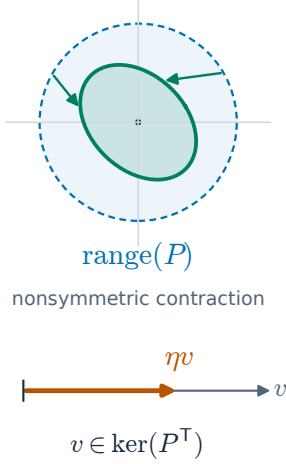
$$\mathbf{I}_r - M^\top M = Q^{-\top} [Q^\top Q - (\mathbf{I}_r - K)^\top (\mathbf{I}_r - K)] Q^{-1} \quad (31)$$

$$= 4Q^{-\top} \text{sym}(K) Q^{-1}, \quad (32)$$

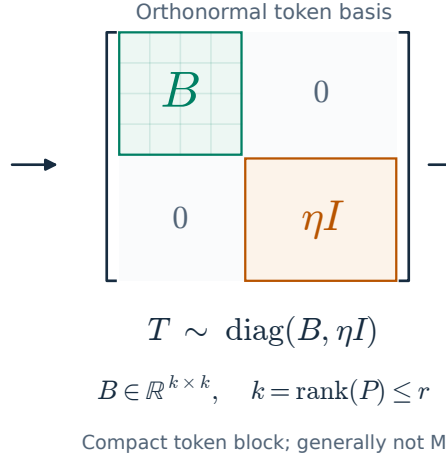
which is (29). It is positive definite, so  $\|M\|_2 < 1$ . The inverse formula maps every strict contraction back to a strictly accretive matrix.  $\square$

Because  $\Omega$  is unrestricted,  $K$  and  $M$  may be nonsymmetric and nonnormal. The chart retains every strict contraction in rank space, and at token level,  $T - T^\top = P(M - M^\top)P^\top$ , so this nonsymmetric structure survives the lift.

**a Token subspaces**



**b Compact representation**



**c Exact gain**

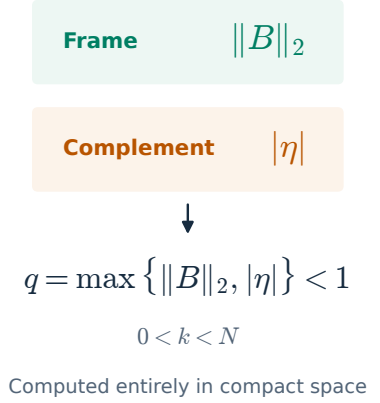


Figure 2: **From token subspaces to an exact gain.** (a) An illustrative nonsymmetric contraction in the frame subspace and scalar multiplication on its orthogonal complement; the scalar sketch uses positive  $\eta$ . (b) In an orthonormal token basis, the realized mixer separates into  $B$  and  $\eta I$ . (c) For  $0 < k < N$ , its norm is the larger of the two block norms. Theorem 4.6 covers either sign of  $\eta$  and the empty and full-rank cases. The ellipse and matrix glyphs are schematic, not trained operator measurements.

### 4.3 Can the realized global gain be computed exactly?

The soft frame identifies the token subspace where contextual mixing acts. Figure 2 separates this compact action from the scalar complement. Because  $P$  is generally not orthonormal, the restricted token block  $B$  below differs from the compact Cayley core  $M$ .

**Theorem 4.6** (Exact realized-gain certificate). *Fix a sample and head, and let  $P$ ,  $M$ , and  $\eta$  be constructed as above. Write the positive eigenspace of  $P^\top P$  as*

$$P^\top P = V_k S^2 V_k^\top, \quad k = \text{rank}(P), \quad (33)$$

where  $S \in \mathbb{R}^{k \times k}$  is positive diagonal, and define

$$B = \eta \mathbf{I}_k + S V_k^\top (M - \eta \mathbf{I}_r) V_k S. \quad (34)$$

Then the singular values of  $T$  are exactly those of  $B$ , together with  $N - k$  copies of  $|\eta|$ . Equivalently,

$$q(X) := \|T\|_2 = \begin{cases} |\eta|, & k = 0, \\ \sigma_{\max}(B), & k = N, \\ \max\{\sigma_{\max}(B), |\eta|\}, & 0 < k < N, \end{cases} \quad q(X) < 1. \quad (35)$$

Moreover,  $\text{rank}(T - \eta \mathbf{I}_N) \leq r$  and  $Tv = \eta v$  for every  $v \in \ker(P^\top)$ .

*Proof sketch.* Let  $Q = P V_k S^{-1}$ . Then  $Q^\top Q = \mathbf{I}_k$ , and substitution gives  $T = Q B Q^\top + \eta(\mathbf{I}_N - Q Q^\top)$ , so an orthogonal basis containing  $Q$  yields the stated singular spectrum. With  $\Delta = (\mathbf{I}_N - P P^\top)^{1/2}$  and  $F = [P \ \Delta]$ ,

$$T = F \text{Diag}(M, \eta \mathbf{I}_N) F^\top, \quad (36)$$

and  $F F^\top = \mathbf{I}_N$ ; hence  $\|T\|_2 < 1$ . Finally,  $T - \eta \mathbf{I}_N = P(M - \eta \mathbf{I}_r) P^\top$ , which gives both the rank bound and  $Tv = \eta v$  on  $\ker(P^\top)$ .  $\square$

The exact certificate requires only an  $r \times r$  Gram eigendecomposition and a singular-value calculation of size at most  $r \times r$ ; no  $N \times N$  kernel is needed. Because the forward uses the input-dependent  $T(X)$  itself, Equation (35) gives the pointwise energy statement

$$\|T(X)C(X)\|_F \leq q(X) \|C(X)\|_F < \|C(X)\|_F \quad (37)$$

whenever  $C(X) \neq 0$ . The guarantee is pointwise in the input-conditioned operator. It does not bound the Jacobian of  $X \mapsto T(X)C(X)$ , which also contains  $(dT)C$ . Along any fixed realized sequence of mixer maps, the gains compose:  $\|T_L \cdots T_1\|_2 \leq \prod_{\ell=1}^L q_\ell$ .

**Remark 4.7** (Why the scalar complement is part of the model). *Without the complement, a conventional form  $\mathbf{I}_N + PAP^\top$  fixes the entire  $\ker(P^\top)$  subspace to gain one. Equation (36) instead assigns that unmodeled subspace the learned per-head scalar  $\eta$ , while retaining the same rank- $r$  correction and the strict contraction certificate. Thus  $T$  is a rank- $r$  correction to an isotropic complement.*

Appendix A proves padding isolation and noncausal permutation equivariance.

## 5 Task Evidence and Realized Operators

We evaluate LSSO as a drop-in replacement for attention in otherwise ordinary bidirectional encoders. All reported LSSO runs use the default DYNAMIC core, Rank-Rotary, rank 32, FP16 automatic mixed precision, and the CUDA implementation. Checkpoints are selected only by validation accuracy, with validation loss breaking an exact tie; the held-out test split is evaluated once after selection. Reported uncertainties are sample standard deviations over seeds 0, 1, and 2. Across the evaluated task configurations, LSSO requires only 1.3%–32.0% of exact MHA’s sequence-dependent mixer-core MACs (Appendix C).

### 5.1 Matched genomic mixer comparison

GenomicBenchmarks collects binary DNA sequence-classification tasks with official train and test partitions [8]. We derive a deterministic stratified validation set only from each training partition. All mixers use the same  $D = 128$ , four-layer, four-head encoder, learned absolute position embeddings, MLP ratio 4, mean readout, optimizer, data order, and 40-epoch budget; only the token mixer changes. Alongside MHA, we test bidirectional ELU + 1 Linear Transformer [16], FAVOR+ Performer with fixed orthogonal random features [5], and the official bidirectional ReLU/cosine cosFormer formulation [26]. Numerically sensitive attention cores run in FP32 under automatic mixed precision. The effective batch is 128, using two-step accumulation for Mouse Enhancers Ensembl. The three linear baselines match MHA’s parameter count; LSSO is 4.1% smaller on the 200-position tasks.

Table 2: GenomicBenchmarks held-out test accuracy (%), mean  $\pm$  sample standard deviation over three seeds under the matched encoder and training protocol. Bold marks the highest mean in each row.

| Task                     | MHA              | Linear                             | Performer        | cosFormer        | LSSO                               |
|--------------------------|------------------|------------------------------------|------------------|------------------|------------------------------------|
| Coding vs. intergenomic  | 80.34 $\pm$ 1.99 | 78.11 $\pm$ 0.10                   | 78.17 $\pm$ 0.01 | 79.12 $\pm$ 0.05 | <b>83.23 <math>\pm</math> 2.46</b> |
| Human or worm            | 65.87 $\pm$ 0.35 | 65.68 $\pm$ 0.22                   | 65.76 $\pm$ 0.08 | 66.07 $\pm$ 0.20 | <b>81.57 <math>\pm</math> 1.18</b> |
| Mouse enhancers Ensembl  | 71.07 $\pm$ 1.65 | <b>71.63 <math>\pm</math> 0.48</b> | 71.35 $\pm$ 0.48 | 71.35 $\pm$ 0.48 | 70.25 $\pm$ 0.83                   |
| Human enhancers Cohn     | 66.83 $\pm$ 0.17 | 66.89 $\pm$ 0.07                   | 67.16 $\pm$ 0.08 | 67.06 $\pm$ 0.23 | <b>67.24 <math>\pm</math> 0.26</b> |
| Human enhancers Ensembl  | 76.37 $\pm$ 0.20 | 75.16 $\pm$ 0.57                   | 74.32 $\pm$ 0.23 | 77.34 $\pm$ 1.53 | <b>78.91 <math>\pm</math> 0.05</b> |
| Human regulatory         | 92.14 $\pm$ 0.05 | 91.79 $\pm$ 0.15                   | 90.10 $\pm$ 0.34 | 92.65 $\pm$ 0.14 | <b>93.10 <math>\pm</math> 0.02</b> |
| Human non-TATA promoters | 84.71 $\pm$ 0.15 | 80.44 $\pm$ 1.84                   | 82.51 $\pm$ 1.07 | 84.42 $\pm$ 0.48 | <b>85.41 <math>\pm</math> 0.17</b> |
| Human OCR Ensembl        | 65.61 $\pm$ 0.54 | 65.12 $\pm$ 0.26                   | 65.36 $\pm$ 0.27 | 66.56 $\pm$ 0.07 | <b>68.55 <math>\pm</math> 0.31</b> |
| Unweighted task macro    | 75.37            | 74.35                              | 74.34            | 75.57            | <b>78.53</b>                       |

Table 2 shows that LSSO has the highest mean on seven of eight tasks; this is not a claim of statistical significance on each task. Its 78.53 macro average exceeds cosFormer, the strongest alternative at 75.57, by 2.96 points and MHA’s 75.37 by 3.16 points. The three linear-attention controls indicate that neither removing softmax nor obtaining linear sequence scaling alone reproduces LSSO’s gain. The macro result first

averages seeds within each task and then gives all tasks equal weight. The largest gain occurs on Human or Worm; Mouse Enhancers Ensembl is the sole exception, where Linear Transformer is highest.

## 5.2 Scope of mechanism attribution

The matched comparison tests the complete LSSO layer. It does not isolate the benefits of the dynamic drive, skew coordinates, rank, or Rank-Rotary. In particular, STATIC removes the dynamic drive to  $K$  while leaving  $P$  input-dependent; it is not a fully input-independent mixer. Matched DYNAMIC/STATIC/ZERO training and core-activity measurements are needed to attribute a task advantage to the generated core. No such ablation results are included here.

## 5.3 Certificates in trained models

We next measure how much certified margin trained models retain. Across three six-layer LRA Text checkpoints and matched held-out token-stream probes from 1K to 8K, all 9,216 head-level observations are finite and satisfy their certified conditions (Table 3). The worst realized gain is 0.9087, leaving at least 0.0913 contraction slack, and the minimum monotonicity margin is 1.1667. Solved states use at most 93.3% of their certified bound, while deterministic adjoint probes use at most 88.1%. Appendix B reports the per-layer distributions and stress-test protocol.

Table 3: Worst observation in the trained-model certificate audit. Bound usage is the measured ratio normalized by its certified upper bound  $1/\mu$ .

| Quantity                  | Certified condition                     | Worst observed         |
|---------------------------|-----------------------------------------|------------------------|
| Realized gain             | $q(X) < 1$                              | $\max q = 0.9087$      |
| Contraction slack         | $1 - q(X) > 0$                          | $\min(1 - q) = 0.0913$ |
| Monotonicity margin       | $\mu > 1$                               | $\min \mu = 1.1667$    |
| Solved-state bound usage  | $\mu \ U^*\ _F / \ Z\ _F \leq 1$        | 0.9329                 |
| Adjoint-probe bound usage | $\mu \ A_K^{-T} G\ _F / \ G\ _F \leq 1$ | 0.8814                 |

The gain in these probes is dominated by the scalar complement  $|\eta|$  (Appendix B). The audit therefore establishes realized operator margins, not the task contribution of the dynamic core. The 8K probes are operator stress tests on assembled token streams, not an 8K classification or length-generalization result.

## 5.4 Long Range Arena context

Long Range Arena (LRA) tests structured reasoning and long-context classification [29]. Following established reporting practice [10], Table 4 quotes results from the original benchmark and subsequent work and groups models by their primary mixing mechanism. General-purpose global mixers and efficient-attention approximations form the first block; structured recurrence and state-space models form the second. This distinction matters because explicit locality and positional structure materially affect LRA performance [24].

We evaluate LSSO on four LRA tasks using six-layer encoders, with  $D = 128$  for ListOps and Retrieval and  $D = 256$  for Text and Pathfinder-32. ListOps, Text, and Retrieval use learned one-dimensional absolute position embeddings with mean pooling; Pathfinder-32 uses factorized row/column embeddings and mean-maximum pooling. Their maximum sequence lengths are 2048, 4096, 4000, and 1024, respectively.

Table 4: LRA test accuracy (%) quoted from the cited sources under their reported protocols. The four-task average is unweighted over the displayed tasks. For LSSO, task scores and the per-seed four-task average are reported as mean  $\pm$  sample standard deviation across three seeds. Bold marks the highest reported mean within each mechanism block.

| Model and source                                                            | ListOps                            | Text                               | Retrieval                          | Pathfinder                         | 4-task Avg.                        |
|-----------------------------------------------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| <i>General-purpose global mixers and efficient-attention approximations</i> |                                    |                                    |                                    |                                    |                                    |
| Transformer [29]                                                            | 36.37                              | 64.27                              | 57.46                              | 71.40                              | 57.38                              |
| Linear Transformer [29]                                                     | 16.13                              | <b>65.90</b>                       | 53.09                              | 75.30                              | 52.61                              |
| Performer [29]                                                              | 18.01                              | 65.40                              | 53.82                              | 77.05                              | 53.57                              |
| Linformer [29]                                                              | 35.70                              | 53.94                              | 52.27                              | 76.34                              | 54.56                              |
| Reformer [29]                                                               | 37.27                              | 56.10                              | 53.40                              | 68.50                              | 53.82                              |
| Luna-256 [22]                                                               | <b>37.98</b>                       | 65.78                              | 79.56                              | 78.55                              | 65.47                              |
| FNet [19]                                                                   | 35.33                              | 65.11                              | 59.61                              | 77.80                              | 59.46                              |
| Nyströmformer [35]                                                          | 37.15                              | 65.52                              | 79.56                              | 70.94                              | 63.29                              |
| <b>LSSO, three seeds</b>                                                    | <b>37.60 <math>\pm</math> 0.65</b> | <b>65.19 <math>\pm</math> 0.46</b> | <b>82.88 <math>\pm</math> 0.08</b> | <b>78.79 <math>\pm</math> 0.34</b> | <b>66.11 <math>\pm</math> 0.24</b> |
| <i>Models centered on structured recurrence or state-space dynamics</i>     |                                    |                                    |                                    |                                    |                                    |
| S4, original [10]                                                           | 58.35                              | 76.02                              | 87.09                              | 86.05                              | 76.88                              |
| S4-LegS, updated [11]                                                       | 59.60 $\pm$ 0.07                   | 86.82 $\pm$ 0.13                   | 90.90 $\pm$ 0.15                   | 94.20 $\pm$ 0.25                   | 82.88                              |
| Mega [23]                                                                   | <b>63.14</b>                       | <b>90.43</b>                       | <b>91.25</b>                       | <b>96.01</b>                       | <b>85.21</b>                       |

Within the global-mixer block, LSSO gives the highest Retrieval, Pathfinder, and four-task average; it is within 0.38 points of the highest ListOps result and 0.71 points of the highest Text result. The same mixer spans symbolic, byte-level, paired-retrieval, and two-dimensional path tasks.

## 6 Execution Cost of the Solved Operator

The default DYNAMIC core with Rank-Rotary has a native, algebraically equivalent CUDA path for inference and first-order training. It avoids both the  $N \times N$  token matrix and the compact Cayley matrix; compact factorization and tiled solves evaluate the equilibrium directly, while the analytic backward reuses saved factors. Mixed-precision outputs and gradients are validated against the FP64 reference. Appendix D gives the execution schedule, workspace, VJP, numerical envelope, and build coverage.

The following measurements use the historical contract-6 implementation; they have not been rerun under the current contract-8 mixed-precision backend. Table 5 measures the complete mixer boundary, including input and output projections. At  $N = 4096$ , native LSSO reaches  $1.79\times$  the Flash-SDPA MHA forward throughput and  $2.30\times$  its training throughput. Across  $N = 1024\text{--}4096$ , forward-backward latency grows by  $3.88\times$  for LSSO and  $7.92\times$  for MHA. These finite-shape timings illustrate the scaling trend; they are not an asymptotic fit or an end-to-end model benchmark.

Table 5: Synchronized milliseconds per batch on an RTX 5070 Ti (lower is better), with  $B = 64$ ,  $D = 256$ ,  $H = 8$ , FP16 AMP, and  $r = 32$  for LSSO (historical contract 6). Ratios above one favor native LSSO; forward-backward includes loss and gradient reset. Bold marks the lowest latency in each row.

| $N$                                         | PyTorch ref. | Native        | MHA          | Ref./native    | MHA/native    |
|---------------------------------------------|--------------|---------------|--------------|----------------|---------------|
| <i>Inference forward</i>                    |              |               |              |                |               |
| 1024                                        | 110.216      | 2.539         | <b>2.131</b> | 43.41 $\times$ | 0.84 $\times$ |
| 2048                                        | 133.940      | <b>5.577</b>  | 5.888        | 24.02 $\times$ | 1.06 $\times$ |
| 4096                                        | 169.549      | <b>10.135</b> | 18.173       | 16.73 $\times$ | 1.79 $\times$ |
| <i>Forward-backward with gradient reset</i> |              |               |              |                |               |
| 1024                                        | 118.858      | <b>7.512</b>  | 8.487        | 15.82 $\times$ | 1.13 $\times$ |
| 2048                                        | 153.955      | <b>15.260</b> | 22.940       | 10.09 $\times$ | 1.50 $\times$ |
| 4096                                        | 211.268      | <b>29.160</b> | 67.195       | 7.25 $\times$  | 2.30 $\times$ |

## 7 Visual Representation and Dense Prediction

### 7.1 ImageNet classification

### 7.2 COCO detection and instance segmentation

### 7.3 ADE20K semantic segmentation

## 8 Limitations

LSSO is designed for bidirectional tasks: its compact statistic uses the full valid sequence. It has no causal masking or autoregressive decoding contract. The token-space family is a rank- $r$  correction to a scalar complement, even though the compact chart covers every real strict contraction. Its realized gain certificate does not bound the end-to-end Jacobian, which also includes derivatives of the input-conditioned operator, feature projections, and the surrounding network.

The task evidence evaluates the complete layer on genomic classification and four LRA tasks. Mechanism attribution, matched comparisons with recent bidirectional linear attention and TTT, and visual transfer remain open. The native packages support  $r \in \{16, 32, 48, 64\}$ , but the reported task runs use  $r = 32$ . Timings are hardware- and shape-specific and use the historical implementation contract; broader hardware, rank, and current-backend measurements are needed to extend those empirical conclusions.

## 9 Conclusion

Complete context permits a sequence layer to construct its mixing state jointly from all tokens. LSSO realizes this choice through a sample-generated compact system and one direct solve. The accretive core, soft frame, and scalar complement jointly specify an expressive compact action, a structured global lift, and an exactly computable realized gain. The resulting algebra supports linear sequence scaling at fixed rank and an equivalent native backward. Matched genomic results establish task value for the complete layer, while long-sequence diagnostics and timings characterize its realized behavior. This construction makes the full-context operator, its geometry, and its execution part of the same design.

## A Complete Theoretical Details

### A.1 Soft-frame spectrum and coverage

Let  $A = U\Sigma V^\top$  be an SVD of the normalized relation matrix. From the positive-diagonal QR factorization in (5),

$$R_{\text{qr}}^\top R_{\text{qr}} = \mathbf{I}_r + A^\top A, \quad P = AR_{\text{qr}}^{-1}. \quad (38)$$

Hence

$$PP^\top = A(\mathbf{I}_r + A^\top A)^{-1}A^\top = U \text{Diag}\left(\frac{\sigma_i(A)^2}{1 + \sigma_i(A)^2}\right) U^\top. \quad (39)$$

This proves (24), including rank-deficient  $A$  and  $N < r$ . The Cholesky/QR coordinates may differ from the symmetric-square-root form  $A(\mathbf{I}_r + A^\top A)^{-1/2}$  by a right orthogonal factor, while their singular spectra and left Gram matrices agree.

Take any  $P \in \mathbb{R}^{N \times r}$  with  $P^\top P \prec \mathbf{I}_r$  and write  $S = \mathbf{I}_r - P^\top P \succ 0$ . Let  $C$  be the upper-triangular Cholesky factor of  $S$ , so  $S = C^\top C$ , and set  $R_{\text{qr}} = C^{-1}$  and  $A = PR_{\text{qr}}$ . Then

$$\begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix} = \begin{bmatrix} P \\ R_{\text{qr}}^{-1} \end{bmatrix} R_{\text{qr}}. \quad (40)$$

The left factor has orthonormal columns because  $R_{\text{qr}}^{-\top} R_{\text{qr}}^{-1} = S$ . It is therefore the unique positive-diagonal reduced QR factorization and produces the desired upper block  $P$ .

## A.2 Converse Cayley map and solve bound

For any  $M$  with  $\|M\|_2 < 1$ , the matrix  $\mathbf{I}_r + M$  is nonsingular. Define  $K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}$ . Direct substitution gives

$$K + K^\top = 2(\mathbf{I}_r + M)^{-\top}(\mathbf{I}_r - M^\top M)(\mathbf{I}_r + M)^{-1} \succ 0. \quad (41)$$

Thus  $K$  is strictly accretive, and the forward and inverse fractional-linear expressions are algebraic inverses wherever defined.

Let  $\mu$  be defined in (20). For a nonzero  $v$ , the skew part of  $K$  contributes zero to the quadratic form, so

$$\langle (\mathbf{I}_r + K)v, v \rangle = \|v\|_2^2 + \|L^\top v\|_2^2 \geq \mu \|v\|_2^2. \quad (42)$$

Cauchy–Schwarz therefore gives  $\|(\mathbf{I}_r + K)v\|_2 \geq \mu \|v\|_2$ , proving  $\sigma_{\min}(\mathbf{I}_r + K) \geq \mu$  and  $\|(\mathbf{I}_r + K)^{-1}\|_2 \leq \mu^{-1}$ .

## A.3 The direct solve resolves a convergent contextual process

Let  $A_K = \mathbf{I}_r + K$ . The following result relates the direct solve to a continuous contextual process.

**Theorem A.1** (Solved contextual inner process). *For fixed contextual quantities  $K$  and  $Z$ , consider the continuous dynamics*

$$\dot{U}(t) = Z - A_K U(t). \quad (43)$$

*From every initial state  $U(0)$ , the dynamics converge to  $U^* = A_K^{-1}Z$  and obey*

$$\|U(t) - U^*\|_F \leq e^{-\mu t} \|U(0) - U^*\|_F. \quad (44)$$

*The rate lower bound is independent of the skew-symmetric part of  $K$ .*

*Proof sketch.* Write  $E(t) = U(t) - U^*$ . Since  $A_K U^* = Z$ ,  $\dot{E} = -A_K E$ . The skew part vanishes in the Frobenius inner product, so

$$\frac{d}{dt} \frac{1}{2} \|E\|_F^2 = -\langle E, A_K E \rangle_F = -\langle E, \text{sym}(A_K) E \rangle_F \leq -\mu \|E\|_F^2, \quad (45)$$

because  $\text{sym}(A_K) = \mathbf{I}_r + LL^\top \succeq \mu \mathbf{I}_r$ . Gronwall’s inequality gives (44).  $\square$

Appendix A also identifies the direct solution as the infinite-step limit of stable gradient descent on the residual objective. The production backward differentiates this solved state rather than a finite unroll.

## A.4 Residual optimization, implicit derivative, and sensitivity

The solved state also uniquely minimizes

$$J(U) = \frac{1}{2} \|A_K U - Z\|_F^2, \quad A_K = \mathbf{I}_r + K, \quad (46)$$

whose Hessian is  $A_K^\top A_K \succ 0$ . Gradient descent with  $0 < \alpha < 2/\sigma_{\max}(A_K)^2$  obeys

$$\|U_j - U^*\|_F \leq \rho^j \|U_0 - U^*\|_F, \quad \rho = \max_i |1 - \alpha \sigma_i(A_K)^2| < 1. \quad (47)$$

Hence the direct solve is both the infinite-time limit in Theorem A.1 and the infinite-step limit of every such residual-gradient run.

Differentiating  $A_K U^* = Z$  gives

$$dU^* = A_K^{-1}(dZ - dA_K U^*). \quad (48)$$

If  $G = \partial \mathcal{L} / \partial U^*$  and  $V = A_K^{-\top} G$ , trace cyclicity yields the exact implicit VJP

$$\frac{\partial \mathcal{L}}{\partial Z} = V, \quad \frac{\partial \mathcal{L}}{\partial A_K} = -V U^{*\top}. \quad (49)$$

For two generated problems, the resolvent identity gives

$$U^\star - U^{\star'} = A_K^{-1}(Z - Z') + A_K^{-1}(K' - K)A_K'^{-1}Z', \quad (50)$$

which proves (22). Taking its differential gives (48) and the stated local bound. The solve and adjoint bounds above further give

$$\|U^\star\|_F \leq \mu^{-1} \|Z\|_F, \quad \|V\|_F \leq \mu^{-1} \|G\|_F, \quad \|dU^\star\|_F \leq \mu^{-1} \|dZ\|_F + \mu^{-2} \|dK\|_2 \|Z\|_F. \quad (51)$$

The margin  $\mu$  certifies the solve and flow rate, while the realized-gain certificate separately captures the effect of skew coordinates. For example, with  $L = \mathbf{I}_2$  and  $\Omega = a \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ ,  $\mu = 2$  for every  $a$ , while the associated  $\|M\|_2$  approaches one as  $|a| \rightarrow \infty$ . This is why the realized token gain is computed separately by (35).

## A.5 Exact compact token spectrum

With the notation of Theorem 4.6,  $Q = PV_k S^{-1}$  has orthonormal columns. Substituting  $P = QSV_k^\top$  into (15) gives

$$T = QBQ^\top + \eta(\mathbf{I}_N - QQ^\top). \quad (52)$$

If  $Q_\perp$  completes  $Q$  to an orthogonal matrix, then for  $k > 0$

$$\begin{bmatrix} Q & Q_\perp \end{bmatrix}^\top T \begin{bmatrix} Q & Q_\perp \end{bmatrix} = \begin{bmatrix} B & 0 \\ 0 & \eta \mathbf{I}_{N-k} \end{bmatrix}. \quad (53)$$

This proves the exact singular-spectrum union and the nonzero-rank cases of (35). The case distinction matters when  $N < r$  and  $P$  has full row rank: then  $k = N$ , there is no token-space complement, and  $|\eta|$  contributes no complement singular value. Since  $B$  may be nonnormal, its largest singular value, not its spectral radius, is the certificate. If  $k = 0$ , then  $P = 0$  and  $T = \eta \mathbf{I}_N$ , giving the first case directly.

## A.6 Padding isolation and permutation equivariance

If an invalid row has  $A_i = C_i = 0$ , the upper QR block has  $P_i = 0$ , so Equation (12) gives  $Y_i = 0$ . Explicit output masking preserves this value after the feature projection.

Let  $\Pi$  permute token rows together with their mask and position identifiers. All rowwise projections, masking, normalization, and phase rotations commute with  $\Pi$ . Uniqueness of the positive-diagonal reduced QR factorization gives  $P' = \Pi P$ , while  $Z' = P'^\top C' = P^\top C = Z$ . The generated  $K$ ,  $M$ , and  $\eta$  are unchanged, and Equation (15) yields

$$\text{LSSO}(\Pi X, \Pi m, \Pi p) = \Pi \text{LSSO}(X, m, p). \quad (54)$$

## B Certificate Diagnostic Protocol

We use the validation-selected seed-0, seed-1, and seed-2 LRA Text checkpoints. Each is a six-layer, eight-head, rank-32 LSSO encoder trained at maximum length 4096. Sixteen probes are formed by concatenating the held-out validation token stream and taking matched prefixes of length 1024, 2048, 4096, and 8192. Thus a probe at one length is a prefix of the same probe at every longer length. This produces  $3 \times 16 \times 6 \times 8 \times 4 = 9,216$  head-level observations.

At each layer, diagnostics are evaluated in FP64 from the realized compact factors. The exact gain  $q(X)$  is obtained from the compact certificate block in Equation (34); no  $N \times N$  operator is formed. We record the monotonicity margin  $\mu$ , the normalized solved-state ratio  $\|U^\star\|_F / \|Z\|_F$ , and the normalized adjoint ratio

$$\frac{\|A_K^{-\top} G\|_F}{\|G\|_F}, \quad A_K = \mathbf{I} + K. \quad (55)$$

Here  $G$  is a deterministic unit-Frobenius Rademacher matrix, shared across lengths for each probe-layer pair, that provides a reproducible probe of the implicit adjoint. Both ratios are checked against their shared upper bound  $1/\mu$ ; equivalently,  $\mu \|U^\star\|_F / \|Z\|_F \leq 1$  and  $\mu \|A_K^{-\top} G\|_F / \|G\|_F \leq 1$  are required for every observation.

We treat 8K as an operator-level stress test. The learned absolute position table is repeated modulo its 4096-position training range, while Rank-Rotary receives centered positions over the full probe. This protocol tests the mixer beyond its training length without introducing a new learned-position extrapolation rule.

Figure 3 shows the corresponding depth and length distributions. The realized-gain curves coincide because the learned scalar complement  $|\eta|$  dominates Equation (35) in these checkpoints, yielding a stable worst-case gain across probe lengths. The compact-core diagnostics remain similarly stable: the monotonicity margin is preserved, the solved-state ratio stays bounded at 8K, and the adjoint-probe ratio remains near 0.50 across depth.

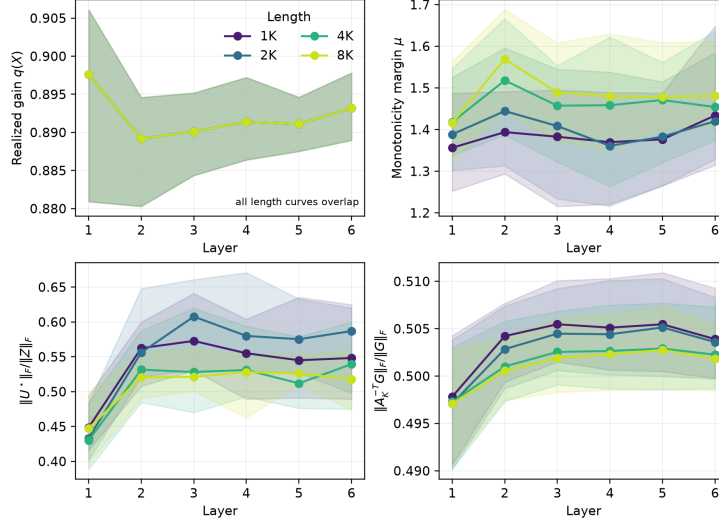


Figure 3: Per-layer certificate diagnostics from three trained LRA Text checkpoints. Curves show medians and shaded 10th–90th percentiles over held-out token-stream probes and heads. All four realized-gain curves overlap; the 8K points are operator-level stress measurements.

## C Mixer-Core MAC Accounting

Table 6 reports the leading sequence-dependent matrix contractions per sample, accumulated over all mixer layers. For depth  $L$ ,  $H$  heads, head width  $d = D/H$ , and rank  $r = 32$ , the accounting is

$$\text{MAC}_{\text{LSSO}} = 2LHNr(r + d), \quad \text{MAC}_{\text{MHA}} = 2LHN^2d. \quad (56)$$

The LSSO count comprises two token–rank frame passes and two frame–content contractions; the MHA count comprises  $QK^\top$  and the attention–value product. Feature projections, elementwise maps, MLPs, task heads, and the  $N$ -independent compact factorizations and solves are excluded from both columns.

Table 6: Sequence-dependent mixer-core GMACs per sample at each task’s maximum length, accumulated over all encoder layers. The final column reports LSSO as a percentage of MHA.

| Task                     | $N$  | $D/L/H$ | LSSO   | MHA     | LSSO/MHA (%) |
|--------------------------|------|---------|--------|---------|--------------|
| <i>GenomicBenchmarks</i> |      |         |        |         |              |
| Coding vs. intergenomic  | 200  | 128/4/4 | 0.0131 | 0.0410  | 32.0         |
| Human or worm            | 200  | 128/4/4 | 0.0131 | 0.0410  | 32.0         |
| Mouse enhancers Ensembl  | 4776 | 128/4/4 | 0.3130 | 23.3576 | 1.3          |
| Human enhancers Cohn     | 500  | 128/4/4 | 0.0328 | 0.2560  | 12.8         |
| Human enhancers Ensembl  | 573  | 128/4/4 | 0.0376 | 0.3362  | 11.2         |
| Human regulatory         | 802  | 128/4/4 | 0.0526 | 0.6586  | 8.0          |
| Human non-TATA promoters | 251  | 128/4/4 | 0.0164 | 0.0645  | 25.5         |
| Human OCR Ensembl        | 593  | 128/4/4 | 0.0389 | 0.3601  | 10.8         |
| <i>Long Range Arena</i>  |      |         |        |         |              |
| ListOps                  | 2048 | 128/6/8 | 0.3020 | 6.4425  | 4.7          |
| Text                     | 4096 | 256/6/8 | 0.8053 | 51.5396 | 1.6          |
| Retrieval                | 4000 | 128/6/8 | 0.5898 | 24.5760 | 2.4          |
| Pathfinder-32            | 1024 | 256/6/4 | 0.1510 | 3.2212  | 4.7          |

## D CUDA Implementation Details

### D.1 Execution contract

The native backend implements DYNAMIC with Rank-Rotary for inference and first-order training and is algebraically equivalent to Section 3. It accepts FP16 or FP32 activations with FP32 learned parameters. The STATIC, ZERO, and Rank-Rotary-off ablations remain reference-only.

The joint input projection in Equation (3) produces contiguous FP32 relation and content coordinates with TF32 enabled on supporting hardware. The separate output projection uses FP16 Tensor-Core operands with FP32 accumulation before casting back to the activation dtype.

The schedule supports

$$r \in \{16, 32, 48, 64\} \quad (57)$$

and any positive head width  $d$  within allocation and launch limits, including  $r > d$ . Runtime right-hand-side panels have width 32 with zero-filled tails.

Boolean validity masks have shape  $[B, N]$ , and position identifiers have shape  $[N]$  or  $[B, N]$ . Invalid projected rows are zeroed before every compact statistic, all sample-dependent normalizations use  $\sqrt{n_{\text{valid}}}$ , and invalid output rows are zeroed again. This covers gapped padding and entirely masked samples in one generic schedule. Position identifiers and valid-token counts are metadata rather than differentiable inputs.

### D.2 Lowering to tiled kernels

The forward pass begins with a rank-phase schedule that constructs the Rank-Rotary sine and cosine table. Consecutive default positions are cached by device, rank, and sequence length; explicit positions use an invocation-local table. One relation-input kernel combines phase rotation,  $1/\sqrt{n_{\text{valid}}}$  normalization, detached common scaling, and storage of the scaled relation coordinates.

The soft frame uses the Gram–Cholesky form algebraically equivalent to the positive-diagonal reduced QR in Equation (5). Short sequences use a single-block FP32 Gram reduction. At 32 or more 32-token tiles, independent partial Grams are reduced in ascending token order before the same Cholesky factorization. Tiled triangular solves then materialize the frame. The cross-moment  $Z = P^T C$  is produced in independent token and right-hand-side tiles using FP16 Tensor-Core multiplicands and FP32 accumulation; no more than 64 token partials are live at once.

A compact-core kernel evaluates

$$\Theta = \Theta_0 + \frac{ZW_{\text{drive}}}{\sqrt{n_{\text{valid}}}}, \quad (58)$$

constructs  $L$  and  $\Omega$ , forms  $LL^T$  using IEEE FP32 FMA, assembles  $\mathbf{I} + K$ , and performs partial-pivot LU factorization. Separate 32-column FP32 solve panels compute  $U^* = (\mathbf{I} + K)^{-1}Z$ . The final tiled kernel

directly evaluates

$$Y = \eta C + P[2U^* - (1 + \eta)Z]. \quad (59)$$

Thus neither the Cayley matrix  $M$  nor a token-space transition matrix is stored.

### D.3 Analytic backward and activation storage

Training records a private contiguous FP32 tape per sample and head together with one-based INT32 LU pivots. The backward reuses the saved LU factors to apply the transposed equilibrium solve, then evaluates analytic VJPs for the dynamic core, accretive parameterization, compact cross-moment, soft frame, and Rank-Rotary transform. It does not differentiate through an unrolled solver. The token-level frame and relation derivatives are fused, so the single-consumer frame adjoint remains on chip rather than making a global  $[BH, N, r]$  round trip. The complement reduction reconstructs its frame-content term in FP32 within an existing compact kernel. Buffers are reused after their last consumer: the compact-core matrix adjoint becomes the frame adjoint, and compact-state partial storage is overwritten after its reduction.

Let  $S = BH$ ,  $T_{32} = \lceil N/32 \rceil$ ,  $T_{64} = \lceil N/64 \rceil$ ,  $Q = \lceil d/32 \rceil$ , and  $C_T = \min(T_{32}, 64)$ . Excluding packed input, output, returned gradients, Rank-Rotary phase tables, and allocator bookkeeping, the saved training tape contains

$$W_{\text{train}} = S(2Nr + 3r^2 + 2rd + 1) \quad (60)$$

FP32 values and  $Sr$  INT32 pivots. The compact inference workspace contains

$$W_{\text{infer}} = S(Nr + 2r^2 + 2rd + 1) \quad (61)$$

FP32 values, plus  $Sr$  transient INT32 pivots. Rank-Rotary phase storage is additional: a shared table contains  $Nr$  FP32 values, while per-sample positions require  $BNr$ . Default consecutive-position tables live in a bounded persistent cache; explicit position tables are invocation-local.

The long-sequence Gram schedule has the logical live storage

$$W_{\text{Gram}} = ST_{32} \frac{r(r+1)}{2} \quad (62)$$

in FP32. Tape-producing calls with  $r \in \{16, 32, 48\}$  place these partials in the not-yet-materialized frame region; inference and  $r = 64$  training allocate them separately. The cross-moment producer uses

$$W_{\text{cross}} = SQC_T r 32 \quad (63)$$

FP32 values, and its lifetime does not overlap the Gram temporary. Native backward scratch contains

$$W_{\text{backward}} = S(T_{64}rd + T_{64} + 3rd + 2r^2) \quad (64)$$

FP32 values. These formulas describe the principal tape/workspace and the explicitly listed scratch tensors owned by the native mixer; full-module peak memory additionally includes phase storage, feature projections, external autograd state, returned gradients, and allocator caching.

### D.4 Numerical and build coverage

The contract-6 backend used for the reported experiments follows a fixed mixed TF32/FP16-Tensor-Core/FP32 contract. The current source uses contract 8 (BF16 ordinary multiplicands, FP16 phase storage, and FP32 sensitive arithmetic); the reported experiments have not been rerun under that contract. The joint input projection uses TF32 where supported. Selected compact contractions and the readout use FP16 multiplicands with FP32 accumulation. Rank-Rotary phases, the soft-frame Gram and Cholesky factorization, the one-ULP-interiorized complement,  $LL^T$ , LU factorization, and triangular solves remain FP32. This fixed precision partition applies to every supported configuration.

FP64 evaluation of the sole reference operator is the numerical oracle. The CUDA acceptance envelope requires finite outputs and gradients, forward relative  $\ell_2$  error at most  $5 \times 10^{-3}$ , and input- and parameter-gradient relative  $\ell_2$  error at most  $10^{-2}$ . The oracle suite covers all four ranks, non-tile-aligned head widths,

lengths through 4776, the  $N = 4096, D = 256, H = 8, r = 32$  long-sequence setting, parallel Gram reduction, batch-specific positions, gapped masks, entirely masked samples, and agreement between inference and tape-producing forward entry points. Acceptance is defined by these mixed-precision error envelopes.

CUDA 12.8 with cuBLASDx and cuSolverDx from MathDx 25.12.1 builds separate artifacts for

$$\text{SM} \in \{75, 80, 86, 87, 89, 90, 100, 120\}. \quad (65)$$

Each binary records its compiled SM and numerical contract. Physical latency and numerical measurements use the RTX 5070 Ti specified below.

## D.5 Wall-clock protocol

Table 5 was measured under WSL2 on one NVIDIA GeForce RTX 5070 Ti (16 GB, SM120), driver 610.88, PyTorch 2.11.0+cu128, CUDA 12.8, and native contract version 6. Training measurements use training mode, while inference measurements use evaluation mode. Each process pregenerates a contiguous input tensor and enables TF32. LSSO uses the default DYNAMIC core with Rank-Rotary, consecutive positions, and no padding mask. The comparison is PyTorch `nn.MultiheadAttention` with bias, zero dropout, batch-first layout, and `need_weights=False`, using PyTorch’s Flash SDPA forward and backward kernels.

The timed boundary includes both operators’ input and output projections, but excludes residual connections, MLPs, data loading, and optimizer updates. The forward–backward measurement differentiates a squared-mean scalar loss and includes setting the input gradient to `None` every iteration and model parameter gradients to `None` after each iteration. No CUDA Graph or `torch.compile` path is enabled. Every configuration runs 50 untimed warmup iterations, followed by seven repetitions of 100 iterations. Each repetition is bracketed by `torch.cuda.synchronize()`, and the reported number is the median synchronized host wall-clock time per iteration. Batch, width, head count, rank, precision, and statistical protocol are fixed across all sequence lengths and implementations. The benchmark entry point is `benchmarks/benchmark.ball.py`.

## E Figure Sources and Production

The conceptual diagrams are supplied as vector PDFs. Their matrix textures and geometric illustrations are schematic and encode no experimental results. Layout and export review used the Scientific Visualization skill from Scientific Agent Skills [15]; the drawing code and operator content were written for this manuscript. The tensor-flow diagrams in ViT<sup>3</sup> [12] and the semantic color grouping in LION [1] informed the visual organization; no source artwork was reused. The existing certificate plot reports the diagnostic protocol in Appendix B.

## References

- [1] Arshia Afzal, Elias Abad Rocamora, Leyla Naz Candogan, Pol Puigdemont, Francesco Tonin, Yongtao Wu, Mahsa Shoaran, and Volkan Cevher. Linear attention for efficient bidirectional sequence modeling. In *Advances in Neural Information Processing Systems*, volume 38, 2025.
- [2] Alaaeldin Ali, Hugo Touvron, Mathilde Caron, Piotr Bojanowski, Matthijs Douze, Armand Joulin, Ivan Laptev, Natalia Neverova, Gabriel Synnaeve, Jakob Verbeek, and Hervé Jégou. XCiT: Cross-covariance image transformers. In *Advances in Neural Information Processing Systems*, volume 34, pages 20014–20027, 2021.
- [3] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [4] Heinz H. Bauschke and Patrick L. Combettes. *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. CMS Books in Mathematics. Springer, Cham, 2 edition, 2017.
- [5] Krzysztof Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking attention with performers. In *International Conference on Learning Representations*, 2021.
- [6] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems*, volume 35, pages 16344–16359, 2022.

- [7] Tri Dao and Albert Gu. Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality. *arXiv preprint arXiv:2405.21060*, 2024.
- [8] Katarína Grešová, Vlastimil Martinek, David Čechák, Petr Šimeček, and Panagiotis Alexiou. Genomic benchmarks: A collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24(1):25, 2023.
- [9] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Conference on Language Modeling*, 2024.
- [10] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations*, 2022.
- [11] Albert Gu, Isys Johnson, Aman Timalina, Atri Rudra, and Christopher Ré. How to train your HiPPO: State space models with generalized orthogonal basis projections. In *International Conference on Learning Representations*, 2023.
- [12] Dongchen Han, Yining Li, Tianyu Li, Zixuan Cao, Ziming Wang, Jun Song, Yu Cheng, Bo Zheng, and Gao Huang. ViT<sup>3</sup>: Unlocking test-time training in vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [13] Andrew Jaegle, Sebastian Borgeaud, Jean-Baptiste Alayrac, Carl Doersch, Catalin Ionescu, David Ding, Skanda Koppula, Daniel Zoran, Andrew Brock, Evan Shelhamer, et al. Perceiver IO: A general architecture for structured inputs and outputs. In *International Conference on Machine Learning*, 2021.
- [14] Haian Jin, Rundi Wu, Tianyuan Zhang, Ruiqi Gao, Jonathan T. Barron, Noah Snavely, and Aleksander Holynski. ZipMap: Linear-time stateful 3D reconstruction via test-time training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- [15] Timothy Kassis, Vinayak Agarwal, Yuhuan He, Darshil Patel, and Aubrey M. Brueckner. Scientific agent skills: A library of procedural knowledge for research agents. *arXiv preprint arXiv:2609.00065*, 2026.
- [16] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 5156–5165. PMLR, 2020.
- [17] Hyunjik Kim, George Papamakarios, and Andriy Mnih. The lipschitz constant of self-attention. In *International Conference on Machine Learning*, pages 5562–5571, 2021.
- [18] Juho Lee, Yoonho Lee, Jungtaek Kim, Adam R. Kosiorek, Seungjin Choi, and Yee Whye Teh. Set transformer: A framework for attention-based permutation-invariant neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3744–3753. PMLR, 2019.
- [19] James Lee-Thorp, Joshua Ainslie, Ilya Eckstein, and Santiago Ontanon. FNet: Mixing tokens with fourier transforms. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4296–4313. Association for Computational Linguistics, 2022.
- [20] Bo Liu, Rui Wang, Lemeng Wu, Yihao Feng, Peter Stone, and Qiang Liu. Longhorn: State space models are amortized online learners. *arXiv preprint arXiv:2407.14207*, 2024.
- [21] Junchen Liu, Sven Elfle, Or Litany, Zan Gojcic, and Ruilong Li. Test-time training with KV binding is secretly linear attention. *arXiv preprint arXiv:2602.21204*, 2026.
- [22] Xuezhe Ma, Xiang Kong, Sinong Wang, Chunting Zhou, Jonathan May, Hao Ma, and Luke Zettlemoyer. Luna: Linear unified nested attention. In *Advances in Neural Information Processing Systems*, volume 34, 2021.
- [23] Xuezhe Ma, Chunting Zhou, Xiang Kong, Junxian He, Liangke Gui, Graham Neubig, Jonathan May, and Luke Zettlemoyer. Mega: Moving average equipped gated attention. In *International Conference on Learning Representations*, 2023.
- [24] Pablo Miralles-González, Javier Huertas-Tato, Alejandro Martín, and David Camacho. On the locality bias and results in the long range arena. *arXiv preprint arXiv:2501.14850*, 2025.
- [25] Hao Peng, Jungo Kasai, Nikolaos Pappas, Dani Yogatama, Zhaofeng Wu, Lingpeng Kong, Roy Schwartz, and Noah A. Smith. ABC: Attention with bounded-memory control. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7469–7483, Dublin, Ireland, 2022. Association for Computational Linguistics.
- [26] Zhen Qin, Weixuan Sun, Hui Deng, Dongxu Li, Yunshen Wei, Baohong Lv, Junjie Yan, Lingpeng Kong, and Yiran Zhong. cosFormer: Rethinking softmax in attention. In *International Conference on Learning Representations*, 2022.

- [27] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning*, 2021.
- [28] Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to Learn at test time: RNNs with expressive hidden states. *arXiv preprint arXiv:2407.04620*, 2025.
- [29] Yi Tay, Mostafa Dehghani, Samira Abnar, Yikang Shen, Dara Bahri, Philip Pham, Jinfeng Rao, Liu Yang, Sebastian Ruder, and Donald Metzler. Long range arena: A benchmark for efficient transformers. In *International Conference on Learning Representations*, 2021.
- [30] Johannes von Oswald, Nino Scherrer, Seijin Kobayashi, Luca Versari, Songlin Yang, Sarthak Mittal, Maximilian Schlegel, Kaitlin Maile, Yanick Schimpf, Oliver Sieberling, Alexander Meulemans, Rif A. Saurous, Guillaume Lajoie, Charlotte Frenkel, Razvan Pascanu, Blaise Agüera y Arcas, and João Sacramento. MesaNet: Sequence modeling by locally optimal test-time training. In *International Conference on Learning Representations*, 2026.
- [31] Ke Alexander Wang, Jiaxin Shi, and Emily B. Fox. Test-time regression: A unifying framework for designing sequence models with associative memory. *arXiv preprint arXiv:2501.12352*, 2025.
- [32] Ruigang Wang and Ian R. Manchester. Direct parameterization of lipschitz-bounded deep networks. *arXiv preprint arXiv:2301.11526*, 2023.
- [33] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-attention with linear complexity. *arXiv preprint arXiv:2006.04768*, 2020.
- [34] Ezra Winston and J. Zico Kolter. Monotone operator equilibrium networks. In *Advances in Neural Information Processing Systems*, 2020.
- [35] Yunyang Xiong, Zhanpeng Zeng, Rudrasis Chakraborty, Mingxing Tan, Glenn Fung, Yin Li, and Vikas Singh. Nyströmformer: A nyström-based algorithm for approximating self-attention. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 14138–14148, 2021.